

EXERCISE SHEET 6

GROUP AND RING THEORY

MAIN QUESTIONS - TO BE DONE BEFORE THE SEMINAR

- 6.1.** Let $M = M_1 \oplus M_2$ be the direct sum of simple modules M_1 and M_2 such that $M_1 \not\cong M_2$. Show that the ring $\text{End}_R(M)$ is a direct sum of division rings.
- 6.2.** Let M be a completely reducible module, and let $K \neq M$ be a submodule of M . Show that M/K is completely reducible.
- 6.3.** Let $R = M_n(F)$, for $n \geq 2$, be the $n \times n$ matrix ring over a field F . Show that R as an R -module is completely reducible. Show also that if A is any left ideal in R , then $A = Re$ for some idempotent e in R .

EXTRA QUESTIONS

- 6.4.** (*Third Isomorphism Theorem*) If A and B are R -submodules of an R -module M such that $A \subseteq B$, show that $(M/A)/(B/A) \cong M/B$ as R -modules.
- 6.5.** Let M be a completely reducible module, and let $K \neq M$ be a non-zero submodule of M . Show that K is completely reducible.